

Algebra

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Preface

THESE NOTES follow, and are meant to cement the course *Algebra*, taught at the University of Western Australia by Dr. Gordon Royle in Semester 2 of 2026. Where needed, the content of lectures is supplemented by *Linear Algebra Done Right* by Sheldon Axler.

0.1 What is Algebra?

I DO NOT YET KNOW ENOUGH TO SAY.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every <i>or</i> for all
$\exists$	There exists
$\implies$	Implies
$\in$	In <i>or</i> is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not <i>or</i> without
$\emptyset$	Empty set
$\sim$	Relates to
$\mapsto$	Maps to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$ n $	The absolute value of n
$\square$	It has been proven

Chapter 1

Fields and Rings

Chapter 2

Equivalence Relations

2.1 Building $\mathbb{Q}$ from $\mathbb{Z}$

Chapter 3

Congruence of Integers

3.1 Modular Arithmetic

3.2 Some Number Theory

3.3 The Rings $\mathbb{Z}_m$

Chapter 4

Vector Spaces

4.1 Subspaces

4.2 Span

4.3 Direct Sums

Chapter 5

Linear Maps